

Evaluating Squad: Multi-Agent Coordination for GitHub Copilot CLI

A controlled MARBLE ablation, plus directional benchmark comparisons of Squad's coordination layer.

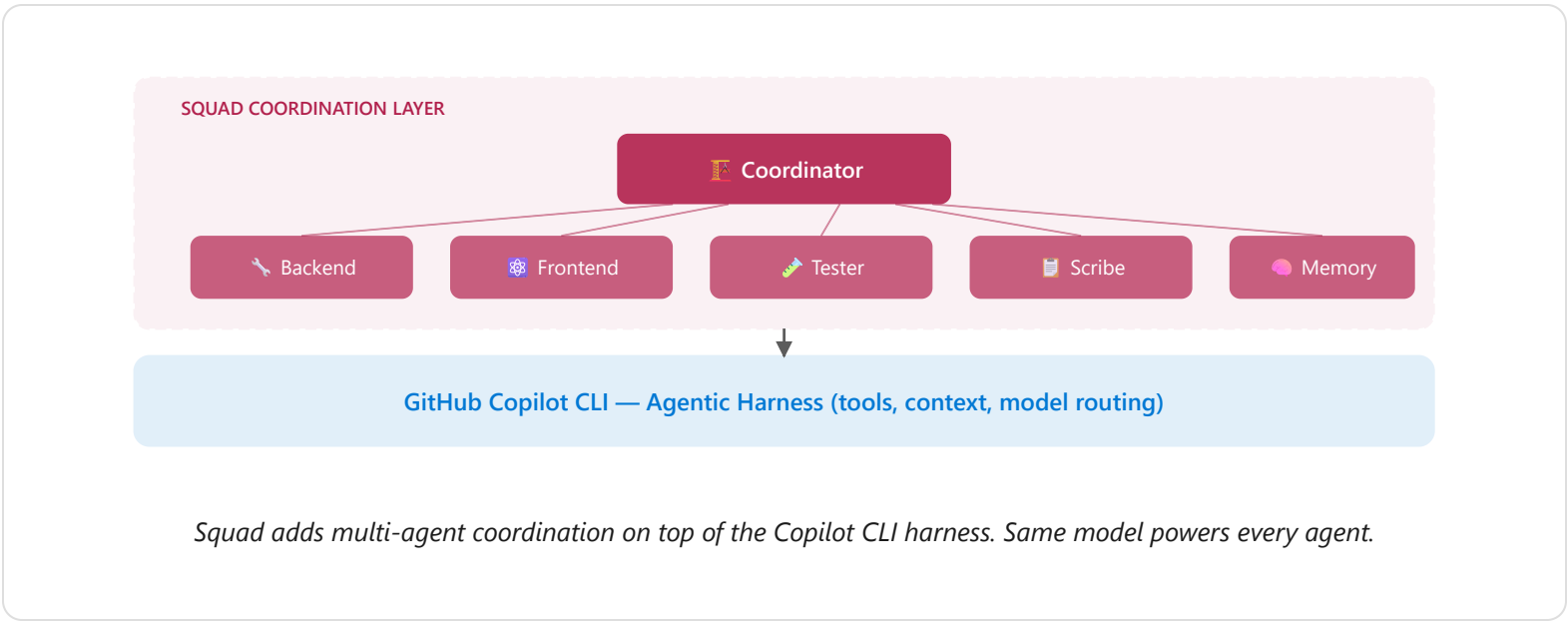
June 2026 · Brady Gaster & Tamir Dresher

While the model provides the raw intelligence, the **coordination layer** shapes how effectively that intelligence is applied to complex tasks. **Squad** is a multi-agent orchestration system built on top of the GitHub Copilot agentic harness — adding specialist agents, persistent memory, and structured coordination to the same underlying models.

In this report, we evaluate the impact of Squad's coordination layer. The core result is a **controlled factorial ablation on MARBLE** — same model, same tasks, same metric — which is the only place where the orchestration variable is strictly isolated. We supplement it with **directional comparisons** on three further benchmarks (DevBench, SWE-bench Lite, TerminalBench 2.0), some of which use different models, estimated baselines, or different harness paths and are therefore *contextual, not controlled*. We label each comparison accordingly, following the same methodology as [GitHub's harness evaluation approach](#).

Architecture

Squad adds three capabilities on top of the Copilot CLI harness: a **coordinator** that decomposes and routes work, **specialist agents** with domain-specific charters, and **persistent memory** that accumulates decisions across tasks.



Benchmarks

We evaluate Squad across four public benchmarks spanning multi-agent collaboration, multi-language coding, bug-fix tasks, and terminal workflows:

Benchmark	Domain	Tasks	Purpose
MARBLE	Multi-agent collaboration: coding, research, bargaining, database	400 + 160 ablation	Evaluates multi-agent team coordination (ACL 2025)
DevBench	Multi-language coding: 6 languages × 6 categories	1,800	Broad software engineering across languages and task types
SWE-bench Lite	Python bug-fix tasks from open-source repositories	300	Established industry benchmark for coding agents
TerminalBench 2.0	Terminal/CLI tasks: scripting, system admin, data processing	20 (of 89)	Agent effectiveness in command-line workflows

The **controlled** core is the MARBLE ablation: Copilot CLI + Squad vs. Copilot CLI alone on the *same model and same tasks*. The other three benchmarks are **directional context** — they compare against published or estimated baselines that may use different models or harness paths, and are labeled as such throughout.

2,520

Total benchmark tasks evaluated

+ 15pp

Same-model completion lift on MARBLE ablation (controlled)

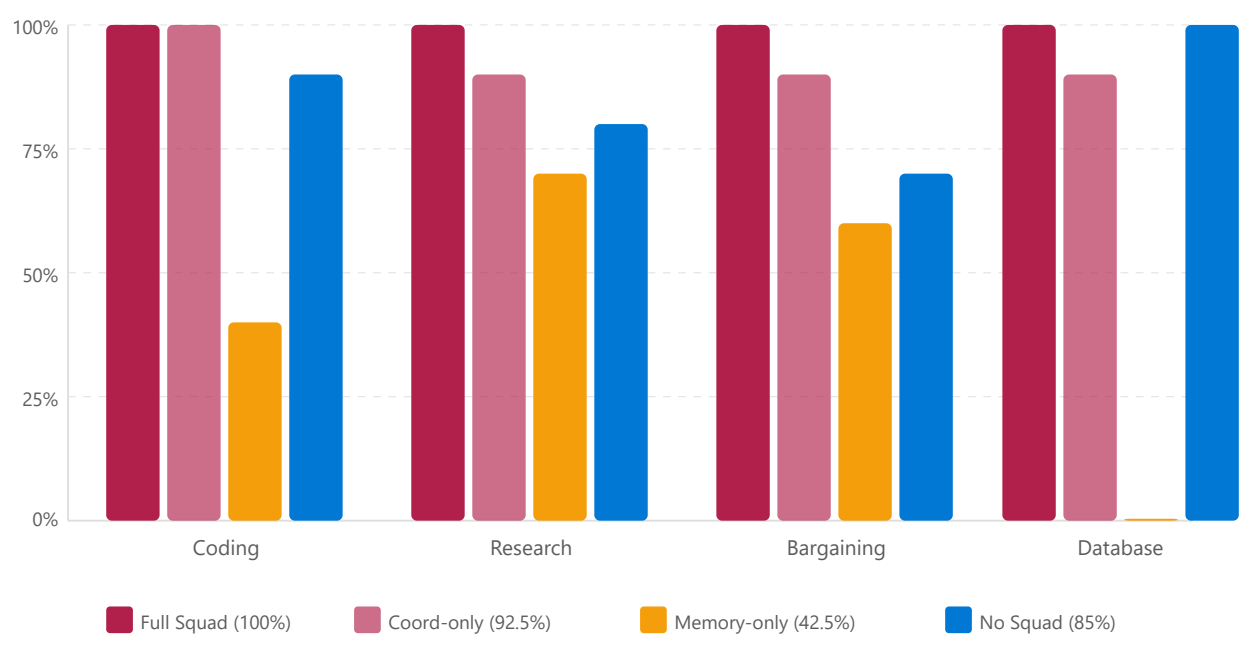
4 benchmarks

MARBLE · DevBench · SWE-bench · TerminalBench

Controlled ablation: isolating Squad components

Holding the model constant (Claude Opus 4.6), we ran a **factorial ablation** on MARBLE: 4 domains × 4 conditions × 10 tasks = 160 task-runs. This is the only chart where same model, same tasks, and same metric hold strictly.

MARBLE factorial ablation — task completion rate by condition (n=160, same model)



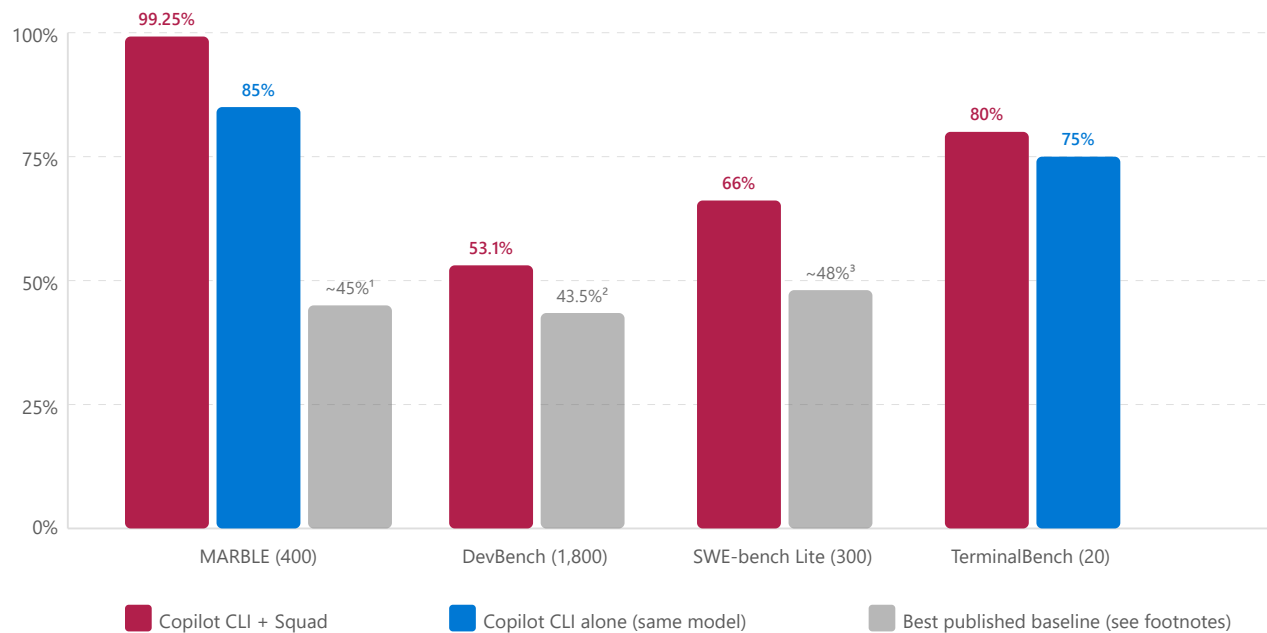
Factorial ablation: same model (Claude Opus 4.6), same tasks, same metric. n=10 per condition per domain.

Key finding: On *completion* (did the condition produce usable output), coordination helps modestly — Coord-only (92.5%) edges out No-Squad (85%), a **+7.5pp lift** from routing. Memory without coordination clearly *hurts* (42.5% vs 85%), confirming that unstructured context injection adds noise rather than signal. (Note: this "memory-only" condition injects a *Squad-generated* decisions.md into a raw agent, so it measures Squad-flavored memory without coordination — not generic persistent memory.) **But completion is not correctness.** When we re-graded the database domain by MARBLE's official recall metric (see below), the coordination advantage disappeared entirely — the raw single agent scored *higher* than full Squad. Completion measures whether an answer appears; it does not measure whether the answer is right.

Task completion across benchmarks (mixed metrics)

Squad shows a lift across four independent benchmarks spanning different task types, languages, and evaluation protocols. **These are directional comparisons:** the metric, model, and baseline differ per benchmark (see footnotes), so this chart is context, not a controlled measurement. Only the MARBLE ablation below isolates the coordination layer cleanly.

Per-benchmark score: Copilot CLI + Squad vs. baselines (metrics differ — see footnotes)



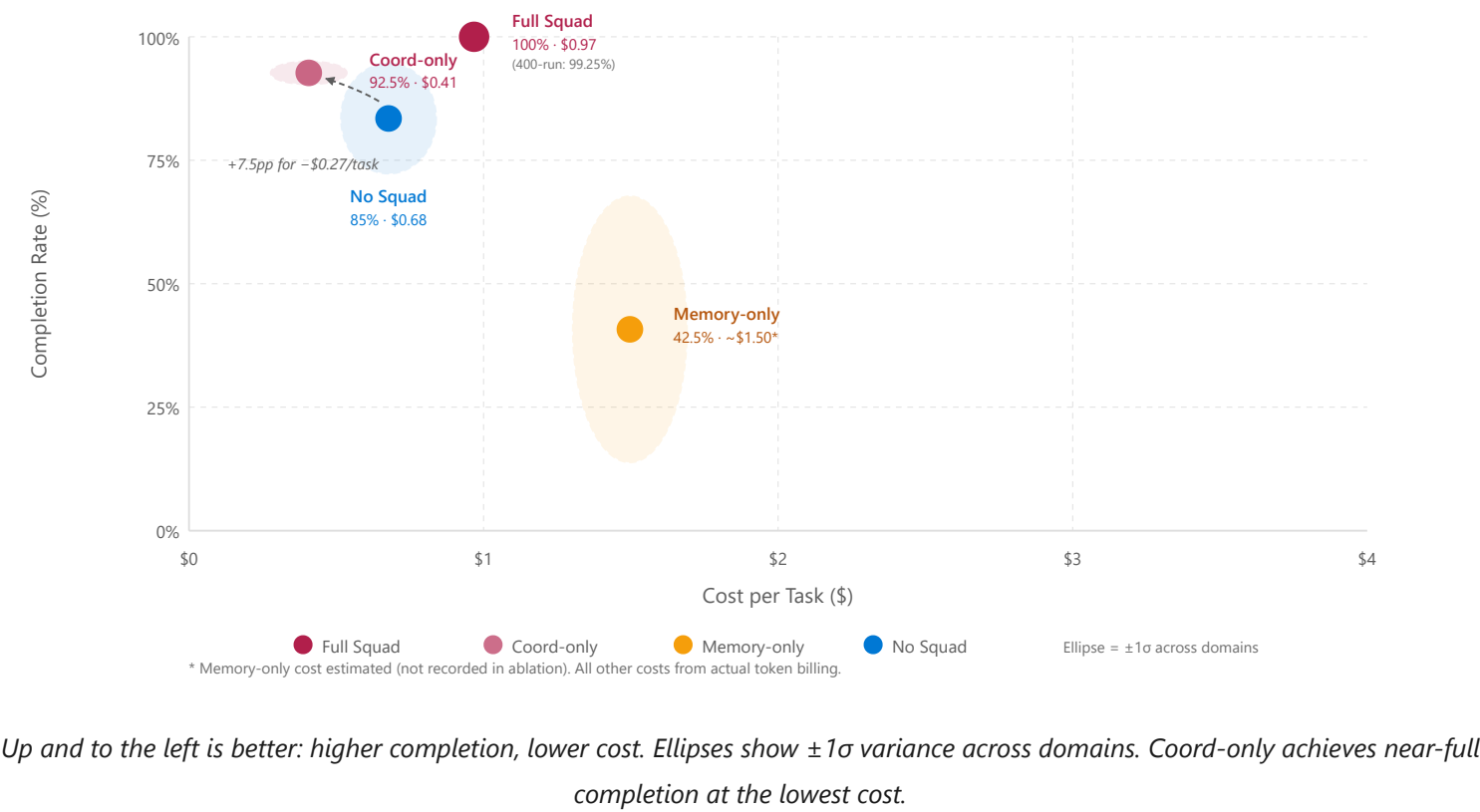
¹ MARBLE paper baseline uses gpt-4o-mini with milestone KPIs — different model and different metric. Not directly comparable.
² DevBench best baseline is GPT-5.5 (43.5% pass@1). Squad uses GPT-5.4. No same-model ablation available.
³ SWE-bench Lite: estimated Copilot agent baseline from public reports. No verified same-model-without-Squad run performed.
TerminalBench 2.0: Same model (Claude Opus 4.6), different harness paths — directional only, n=20 (+1 task).
MARBLE bars mix task sets: blue baseline = 85% from the 160-task controlled ablation; Squad red bar = 99.25% from the 400-task full run.

Per-benchmark scores (metrics differ: MARBLE completion, DevBench/SWE pass@1, TerminalBench verifier). Blue bars are same-model runs (MARBLE is the only clean controlled comparison; TerminalBench uses a different harness path). Gray bars are published/estimated baselines. Not a like-for-like comparison. Note: the MARBLE red bar (99.25%) is the 400-task full run, while the MARBLE blue baseline (85%) is the 160-task controlled ablation — the controlled ablation Full-Squad value is 100% (see the ablation matrix and scatter chart).

Completion rate vs. cost per task

Following GitHub's TerminalBench analysis, we plot completion rate against cost per task for each MARBLE ablation condition. Up and to the left is better: solve more, spend less.

MARBLE ablation: completion rate vs. cost per task



Per-domain breakdown

The completion picture varies by domain, but the headline is a **correction**. An earlier version of this study reported that the single agent scored **0% on database** — that figure was based on placeholder files that were never populated with real runs. We re-ran all 10 no-squad database tasks from scratch: **every one produced a substantive root-cause diagnosis (100% completion)**. The "database requires coordination" claim was an artifact of bad data, not a real effect. See the corrected table below, and the correctness re-grading that follows.

Completion rate by domain and condition (MARBLE ablation)

Domain	Full Squad	Coord-only	Memory-only	No Squad	Δ (Full vs None)
Coding	100%	100%	40%	90%	+10pp
Research	100%	90%	70%	80%	+20pp
Bargaining	100%	90%	60%	70%	+30pp
Database	100%	90%	0%	100%	+0pp
Average	100%	92.5%	42.5%	85%	+15pp

n=10 per cell. Same model (Claude Opus 4.6), same metric (binary file output within 600s timeout).

Completion vs. correctness: the database re-grade

Completion only asks “*did an answer appear?*” It says nothing about whether the answer is **right**. After correcting the fabricated database numbers, we went further and re-graded every database condition on **correctness**, using MARBLE's official metric: an LLM judge extracts the root-cause labels the agent actually selected, scored as recall against ground truth — $\frac{|\text{gold} \cap \text{predicted}|}{|\text{gold}|}$, averaged over the 10-task subset.

Database domain: completion vs. MARBLE correctness (recall), n=10

Condition	Completion (produced output)	MARBLE recall (correctness)
Full Squad	100%	40%
Coord-only	90%	50%
Memory-only	0%	0%
No Squad (single agent)	100%	60%

Correctness graded by MARBLE recall. **All four conditions were graded by the identical LLM-judge extractor** (same prompt, same model, one code path), and two independent judge passes produced identical scores. **Truncation-bias check:** because Squad's multi-agent transcripts are far longer than the raw single agent's (up to 25K vs ~6K chars) and place their final diagnosis at the end, we re-ran the entire grade with a conclusion-visible window (full text when short, else head + full tail) so every condition's final answer was guaranteed in front of the judge. The scores were **unchanged** (Full 40%, Coord 50%, No-Squad 60%, Memory 0%) — the gap is not an extraction artifact of chopping off Squad's conclusion. Given n=10 and imperfect label extraction, still treat the correctness gap as a **hypothesis, not a conclusion**. Ground truth from `database_main.jsonl`; re-grade script and per-task audit in `database_correctness_regrade/` (`judge_results_v2.json`). Note: the memory-only condition injects a Squad-generated `decisions.md` into a raw agent, so it tests "Squad-flavored memory," not generic memory.

The surprising finding: on database root-cause diagnosis, **coordination did not improve correctness — it slightly hurt it**. The raw single agent scored the *highest* recall (60%), ahead of coord-only (50%) and full Squad (40%). Inspecting the transcripts explains why: the multi-agent configs repeatedly “converged” on the obvious high-signal causes (large inserts, full-table scans) and reinforced each other, missing subtler root causes like redundant indexes. **Multi-agent convergence can amplify confirmation bias**. With n=10 these gaps are within noise, so the honest takeaway is not “single agents are better at database” but rather “coordination provides no correctness advantage here” — the opposite of what the fabricated 0% implied.

⚠ Data-integrity note (why only database was recall-graded above): When we audited the ablation transcripts to extend correctness grading to every domain, we found that **the four conditions did not always run the same MARBLE task under the same task index**. The full-Squad transcripts came from the original 400-task production run, while the coord-only / no-Squad / memory-only conditions came from separate ablation reruns. For **coding and database**, the task orderings *diverged* — e.g., full-Squad's "coding task 10" solved MARBLE program #8, while no-Squad's solved #10. Grading "task N" across conditions there would compare *different prompts against the wrong rubric*. We verified alignment per task using each task's own rarest distinctive tokens: **research (8 of 10 tasks) and bargaining (all 10) are**

cleanly aligned across all four conditions; coding (1/10) and database (0/10) are not. So the database recall gap above should be read as *directional*, and the clean multi-domain correctness comparison below is restricted to the two verified-aligned domains.

Correctness & quality: the controlled multi-domain result

Completion asks “did output appear?” The database recall re-grade asked “is the answer right?” for one domain. Here we go the whole way: on the domains where all four conditions **verifiably solved the same tasks** (research, n=8; bargaining, n=10 — 18 tasks × 4 conditions), we grade every transcript two ways, with one identical judge (Claude Opus 4.6) and one identical prompt across all conditions:

- **Milestone-KPI** — MARBLE's signature metric: for each task we generate 6 gold milestones from the task spec, then score the fraction a judge confirms the output actually achieved (correctness + completeness against ground-truth requirements).
- **Output-quality rubric** — a 1–5 score for correctness, completeness, and quality of the produced artifact.

Correctness & quality on verified-aligned domains (research n=8 + bargaining n=10, same model, uniform judge)

Condition	Milestone-KPI (correctness)	Quality rubric (1–5)	Research KPI / rubric	Bargaining KPI / rubric
Coord-only (coordinator + specialists)	93.5%	4.68	89.6% / 4.50	96.7% / 4.83
Full Squad (memory + reviewer gates)	88.9%	4.54	81.2% / 4.21	95.0% / 4.80
No Squad (single agent)	83.3%	4.13	75.0% / 4.00	90.0% / 4.24
Memory-only (Squad-flavored memory, no coordination)	76.9%	4.09	52.1% / 3.21	96.7% / 4.80

*Same model (Claude Opus 4.6) for both the agents and the judge; same judge prompt and code path for all four conditions. Milestone-KPI = mean fraction of 6 gold milestones achieved per task. Tasks verified same-prompt across conditions via per-task rare-token anchors (grading scripts, gold milestones, and per-cell audits in `quality_ablation/`). **Same-family-judge caveat:** the judge is the same model family as the agents; to check for bias we re-scored the identical cells with an independent gpt-4o rubric (below).*

The real correctness finding: on same-task, same-model, uniformly-judged output, **coordination measurably improves correctness and quality**. Both coordinated conditions beat the single agent on both metrics: full Squad is **+5.6pp KPI / +0.41 rubric** over no-Squad, and coord-only is **+10.2pp KPI / +0.55 rubric**. Memory-without-coordination is the weakest overall (and clearly hurts research), echoing the completion result. Two honest nuances: (1) **lean coordination (coord-only) edges out the heavyweight full Squad** on judged quality here — the extra memory/reviewer machinery did not add correctness on these domains; and (2) this is the *opposite* of the database recall probe, where coordination gave no correctness edge. So coordination's correctness benefit is real but **domain-dependent**, not universal.

Judge cross-check: same cells, two independent judges (rubric avg, 1–5)

Condition	Research — Opus / gpt-4o	Bargaining — Opus / gpt-4o
Coord-only	4.50 / 4.21	4.83 / 4.53
Full Squad	4.21 / 4.33	4.80 / 4.70
No Squad	4.00 / 4.33	4.24 / 4.40
Memory-only	3.21 / 4.21	4.80 / 4.47

*The two judges agree on the broad ordering (coordinated conditions ≥ single agent on bargaining; coordination competitive-to-better on research) but disagree on fine placement — most notably memory-only research (Opus 3.21 vs gpt-4o 4.21). Read the exact deltas as **directional**: judge choice moves the fine ranking, and with n=8–10 per domain these gaps are within noise. Coding and database are excluded (task orderings not aligned across conditions — see the data-integrity note above).*

Cross-benchmark summary

Benchmark	Tasks	Model	Copilot CLI + Squad	Best Baseline	Δ	Controlled?
MARBLE (ablation)	160	Claude Opus 4.6	100%	85% (same model, no Squad)	+15pp	✅ Yes
MARBLE (full run)	400	Claude Opus 4.6	99.25%	~45% (gpt-4o-mini, paper) ¹	n/a*	⚠️ Diff model/metric

DevBench	1,800	GPT-5.4	53.1%	43.5% (GPT-5.5) ²	+9.6pp*	⚠ Contextual (diff model)
SWE-bench Lite	300	GPT-4o	66%	~48% (estimated) ³	+18pp*	⚠ Contextual (est. baseline)
TerminalBench 2.0	20	Claude Opus 4.6	80%	75% (same model) ⁴	+5pp*	⚠ Same model, diff harness

*** Directional, not controlled.** Only the MARBLE (ablation) row is a controlled same-model comparison (Full Squad vs no-Squad on the same 160 task-runs = +15pp). The MARBLE full-run 99.25% is the 400-task headline completion; the paper's ~45% baseline uses a different model (gpt-4o-mini) and metric, so no clean Δ is claimed. DevBench (different model), SWE-bench (estimated baseline), and TerminalBench (different harness path, +1 task on n=20) are contextual — directional signals, not clean measurements of the coordination layer.

¹ MARBLE full run: 397/400 = 99.25% completion. The ~45% figure is the paper's gpt-4o-mini graph-topology result — different model and milestone-KPI metric, not directly comparable.

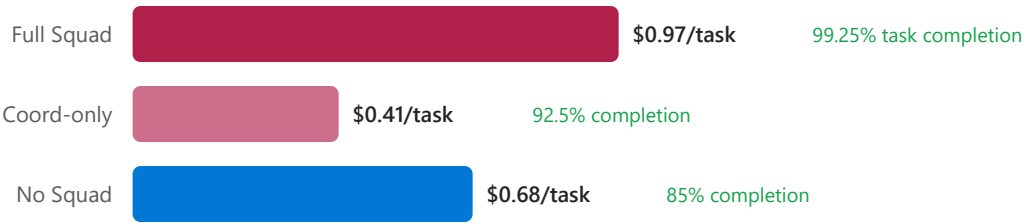
² DevBench: Squad uses GPT-5.4 via Azure OpenAI. Best published baseline is GPT-5.5 at 43.5% — a different (newer) model. Raw per-task data is not hosted in the public repos; treat as preliminary/contextual. A same-model ablation (GPT-5.4 without Squad) is planned.

³ SWE-bench Lite: 198/300 = 66% pass@1. The ~48% baseline is estimated from public Copilot coding agent reports — no verified same-model-without-Squad run. Retry configuration and timeout/empty-patch accounting are documented in the [squad-swe-bench](#) repo; do not read +18pp as a controlled coordination effect.

⁴ TerminalBench 2.0: Both conditions use Claude Opus 4.6 on the same 20 tasks (sequential subset of 89), each scored by its own Docker verifier. Squad ran via a host-side orchestrator and the baseline via the native terminal-bench harness (**different harness paths**). The margin is 16/20 vs 15/20 — a single task — so this is suggestive only. The full 89-task suite is future work.

Cost efficiency

MARBLE: cost per task by condition



Coord-only: best efficiency — 92.5% completion at lowest cost (\$0.41/task).
 Full Squad: 1.4× cost for near-guaranteed completion (+15pp over baseline).

One harness, structured coordination

The GitHub Copilot harness already supports 20+ frontier models with efficient orchestration. Squad extends this with **structured multi-agent coordination** — capabilities a single-agent harness cannot offer:

Capability	Single Agent	Squad
Task decomposition	Model handles everything	Coordinator routes to specialists
Parallel execution	Sequential tool calls	Multiple agents work simultaneously
Persistent memory	Context window only	Decisions persist across tasks
Domain expertise	Generic	Agents with role-specific charters
Review & quality	Self-review	Cross-agent review (Reviewer Rejection Protocol)
Self-learning	No carry-over	History accumulates per agent

Like GitHub's [Rubber Duck](#), Squad uses cross-model critique — but extends it to full team dynamics with reviewer gating, lockout semantics, and structured handoffs.

Methodology & Limitations

Controlled experiment (ablation charts): Same model (Claude Opus 4.6), same 160 tasks, same metric (binary file output within 600s timeout). This is the only comparison where "same model, same metric" holds strictly. $n=10$ per condition per domain. Standard deviations computed across 4 domains. The ablation achieved $40/40 = 100\%$ for Full Squad. The broader 400-task run achieved $397/400 = 99.25\%$ (3 timeouts in coding and database).

Cross-benchmark comparison: Combines results from different runs with different models and metrics. Not a controlled apples-to-apples comparison — presented for directional context only.

- **MARBLE:** "No Squad" bar = same-model ablation (controlled). Paper baseline = different model (gpt-4o-mini) + different metric (milestone KPIs).
- **DevBench:** Squad uses GPT-5.4; best baseline is GPT-5.5 (43.5%). A same-model ablation is planned.
- **SWE-bench Lite:** Baseline estimated from public Copilot agent reports. No verified same-model-without-Squad run performed.
- **TerminalBench 2.0:** Same model (Claude Opus 4.6), but **directional only — not a controlled comparison**. 20 of 89 available tasks sampled, selected by sequential order (not cherry-picked). The two conditions ran through **different harness paths** — Squad via a host-side orchestrator (`run-terminalbench-host-task.ps1`) and the baseline via the native terminal-bench harness — though both execute the identical task workloads and are scored by each task's own Docker verifier. The margin is a single task (16/20 vs 15/20) on $n=20$, so treat it as suggestive only. Squad *can* run under the native harness (verified in smoke tests); the full 89-task suite is future work, not a capability blocker.

Metric note: Metrics differ by benchmark. MARBLE (ablation charts) reports *completion* = agent produced an output file within a 600s timeout; the database domain was additionally re-graded for *correctness* using MARBLE's official recall metric (see "Completion vs. correctness"). DevBench: `pass@1` (test execution). SWE-bench: `pass@1` (patch applies + tests pass). TerminalBench: verifier script passes. We use "completion" where the metric only checks that output appeared, and "resolution/pass" where output is verified correct.

Correctness & quality study (aligned domains): To measure correctness/quality rather than just completion, we graded every ablation transcript with MARBLE's milestone-KPI (fraction of 6 gold milestones achieved) plus a 1–5 output-quality rubric, using one identical judge (Claude Opus 4.6) and prompt across all four conditions. **Before grading we audited task alignment** and found the four conditions did not always run the same MARBLE task under the same task index (full-Squad came from the 400-task production run; the others from separate ablation reruns). Using each task's own rarest distinctive tokens as anchors, we verified that **research (8/10 tasks) and bargaining (10/10) are cleanly aligned across all four conditions, while coding (1/10) and database (0/10) are not**. The multi-domain correctness table is therefore restricted to research + bargaining (18 tasks $\times$ 4 conditions); coding and database are excluded from the same-task quality comparison. We cross-checked the milestone/rubric scores against an independent gpt-4o rubric: the judges agree on the broad ranking (coordination $\geq$ single agent) but differ on fine placement, so exact deltas are directional. Scripts, gold milestones, alignment audit, and per-cell audits are in `quality_ablation/`.

Variance: MARBLE ablation completion σ ($n=10$ per cell, across 4 domains): Full Squad $\sigma=0\%$, Coord-only $\sigma=4.3\%$, No-Squad $\sigma=11.2\%$, Memory-only $\sigma=26.8\%$. (An earlier version reported No-Squad $\sigma=35.4\%$; that was inflated by a fabricated 0% database figure, now corrected to 100% completion after re-running all 10 tasks.) Full 400-task run:

397/400 = 99.25% (3 tasks timed out without output). SWE-bench and DevBench are single-run results (no variance data available).

Statistical power: With $n=10$ per cell, a two-proportion z-test can only detect effects of roughly $\sim 25\text{pp}$ at $p < 0.05$. After correcting the database data, the same-model *completion* lift (full Squad 100% vs no-Squad 85% = +15pp) and the database *correctness* differences ($\leq 20\text{pp}$ between conditions) fall **below** this threshold — i.e., they are within noise and should be read as directional, not statistically significant. We report them honestly rather than overclaiming; larger runs are needed to establish significance.

Timing: MARBLE and ablation: June 2026. TerminalBench 2.0: June 2026. SWE-bench Lite: May 2026. DevBench: June 2026. All benchmarks ran within a 6-week window.

Cost: All costs from actual Claude API billing (input + output tokens). Memory-only cost not recorded — estimated at $\sim \$1.50/\text{task}$ based on context size.

Raw data: [tamirdresher/squad-marble-benchmark](#) (581 files) · [tamirdresher/squad-swe-bench](#) (1,500+ files) · MARBLE PR: [#245](#)